Mariem Nasri

AI & Data Science Engineer
+216 27 582 395 | mariem.nasri773@gmail.com
linkedin.com/in/mariem-nasri-623789298 | github.com/Mariem-Nasri

SUMMARY

AI & Data Science Engineer driven by the challenge of turning ideas into intelligent, real-world solutions. I enjoy exploring how data and AI can solve complex problems, designing practical systems, and transforming innovative concepts into reliable applications. Always curious, eager to learn, and focused on building technology that makes a meaningful difference.

EDUCATION

National Engineering Diploma in Computer Science - Data Science Specialization - ESPRIT 2023 – 2026
Preparatory Cycle in Physics and Chemistry - IPEIN, Preparatory Institute for Engineering Studies of Nabeul 2021 – 2023
Baccalaureate in Experimental Sciences (High Honors) - Hannibal High School 2020 – 2021

PROFESSIONAL EXPERIENCE

VERMEG - Data Science Intern 2026
DocAI: Intelligent Field Extraction from Scanned Financial Agreements

- Developed an end-to-end **document intelligence system** to extract **structured data from scanned financial agreements** using OCR and a **5-agent LLM architecture** for dynamic field extraction.
- **Benchmarked five OCR approaches** and selected **GLM-OCR** (**7.32% CER**, **0.754 TEDS**); designed the multi-agent pipeline with **local Ollama inference** to address data residency and sensitive financial data requirements.
- Built an operational **FastAPI + React** application with **Keycloak authentication**, **Docker** deployment, and **Prometheus**/**Grafana monitoring**, integrating the complete AI pipeline.

FARNESS UAV SOFTWARE - AI Intern 2025
Object Detection & Tracking: Pipeline, Puddle, Leak and Crack Detection with Drones

- Developed a **real-time drone-based computer vision system** for detecting **pipeline cracks, leaks, and puddles** using **YOLOv12**, enabling live infrastructure inspection.
- Achieved **98.9% precision**, **98.4% recall**, and **98.6% mAP@50**; implemented **ByteTrack and SORT** for **real-time multi-object tracking** of detected anomalies.
- Integrated detection and tracking into a **live video pipeline**, enabling continuous identification and monitoring of infrastructure anomalies during drone inspections.

ADDINN Group - Full-Stack Developer Intern 2024
Product Management Web Application

- Developed a **full-stack product management application** using **Angular, .NET 6, and PostgreSQL**, implementing RESTful CRUD operations for product management.
- Designed **secure user authentication** with form validation and password encryption.

PERSONAL PROJECTS

Automated Candidate Screening Agent

- Developed an end-to-end **AI recruitment system** automating **resume-job matching, candidate ranking, and recruiter feedback** using **LLMs and RAG**.
- Built a **semantic matching and ranking engine** to evaluate resumes against job descriptions based on contextual similarity.
- Implemented an **AI feedback agent** using **RAG** to ground structured insights across **Summary, Skills, Experience, and Education** in the matched resume and job description.

FarmWise: AI-Driven Agricultural Solutions

- Developed an **AI-driven agricultural decision-support platform** providing data-driven recommendations for crop management and planning in Tunisia.
- Designed a **hybrid GCN-LSTM model** combining **graph-based learning** to capture relationships between static soil properties and **temporal modeling** to learn patterns from previous crop sequences, enabling crop rotation recommendations.
- Integrated the recommendation model into a **Django web application**, providing real-time crop rotation recommendations through an interactive platform.

TECHNICAL SKILLS

- **Programming:** Python, R, C
- **Machine Learning & Generative AI:** Machine Learning, Deep Learning, NLP, RAG, Computer Vision, OCR, LLMs, Generative AI, AI Agents, Fine-tuning, vLLM, Ollama, Image Classification, Recommendation Systems
- **DevOps & MLOps:** CI/CD, Docker, MLflow, Jenkins, GitHub Actions, Prometheus, Grafana, Model Deployment
- **Web Development:** FastAPI, Django, React.js, Angular, .NET, REST APIs
- **Databases:** PostgreSQL, MySQL, Oracle
- **Languages:** English, French